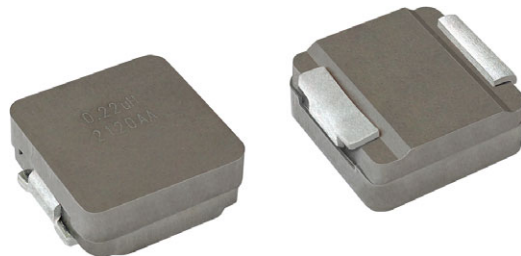


Automotive Inductor, Ultra Low DCR, High Temperature (155 °C) Series



FEATURES

- Magnetically shielded construction
- Optimized for maximum efficiency at high load currents in high frequency converters
- Patented coil design achieves ultra low DCR and robust design
- Thermally conductive structure minimizes hot spots for enhanced heat dissipation over ferrite technologies in natural convection and active cooling environments
- Handles high transient current spikes without saturation
- AEC-Q200 qualified
- IHSR design; PATENT(S): www.vishay.com/patents
- Material categorization: for definitions of compliance please see www.vishay.com/doc?99912

AUTOMOTIVE
GRADE

RoHS
COMPLIANT

HALOGEN
FREE
GREEN
(5-2008)

LINKS TO ADDITIONAL RESOURCES



Product Page



Calculators

APPLICATIONS

- High current load EMI filters (12 V / 100 A or 48 V / 100 A)
- LIDAR boost inductor for laser diode with GaN FETs
- High frequency, low voltage converters (12 V to 1 V) for automotive control units
- LC filter inductor for brushless DC motors (BLDC), electronic power steering, super chargers

STANDARD ELECTRICAL SPECIFICATIONS

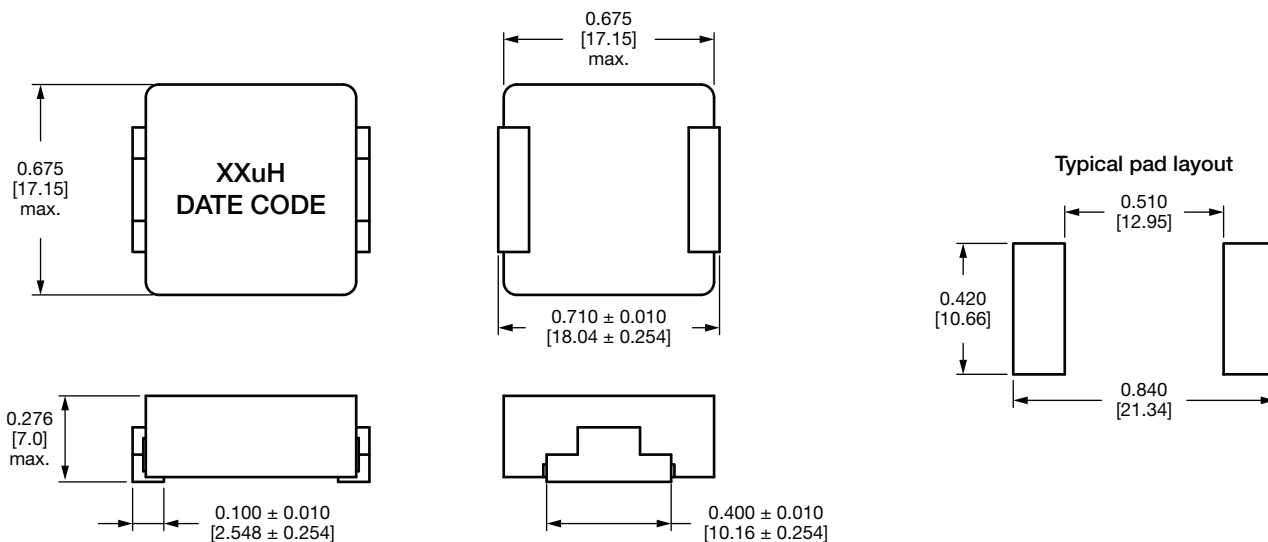
PART NUMBER	L ₀ INDUCTANCE ± 20 % AT 0 A (μH)	DCR AT 25 °C (mΩ)		HEAT RATING CURRENT DC (A)		SATURATION CURRENT DC (A)	
		TYP.	MAX.	TYP. (1)	TYP. (2)	TYP. (3)	TYP. (4)
IHSR6767GZERR22M5A	0.22	0.24	0.26	100	141	107	155

Notes

- All test data is referenced to 25 °C ambient
 - Test condition: 100 kHz, 0.25 V
 - Operating temperature range -55 °C to +155 °C
 - The part temperature (ambient + temp. rise) should not exceed 155 °C under worst case operating conditions. Circuit design, component placement, PCB trace size and thickness, airflow and other cooling provisions all affect the part temperature. Part temperature should be verified in the end application
- (1) DC current (A) that will cause an approximate ΔT of 40 °C
(2) DC current (A) that will cause an approximate ΔT of 80 °C
(3) DC current (A) that will cause L₀ to drop approximately 20 %
(4) DC current (A) that will cause L₀ to drop approximately 30 %

PATENT(S): www.vishay.com/patents

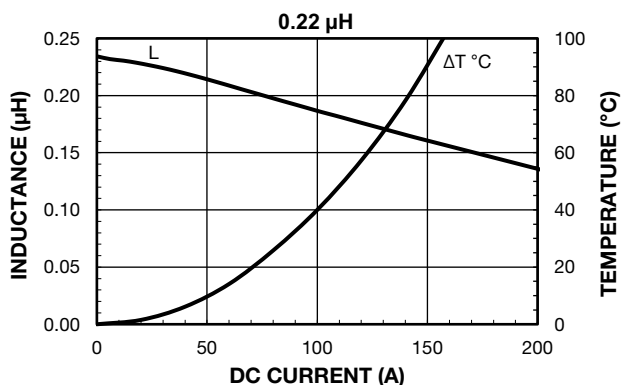
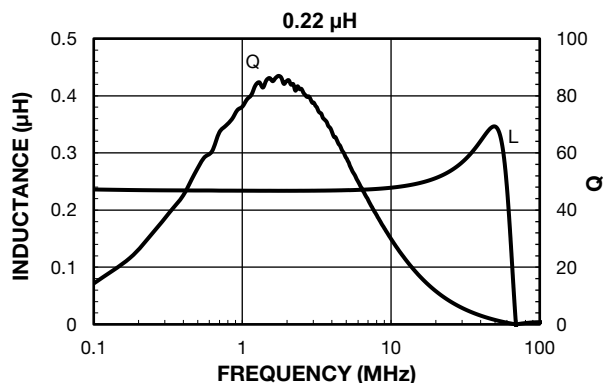
This Vishay product is protected by one or more United States and international patents.

DIMENSIONS in inches [millimeters]

DESCRIPTION

IHSR-6767GZ-5A	0.22 μH	± 20 %	ER	e3
MODEL	INDUCTANCE VALUE	INDUCTANCE TOLERANCE	PACKAGE CODE	JEDEC® LEAD (Pb)-FREE STANDARD

GLOBAL PART NUMBER

I H S R	6 7 6 7 G Z	E R	R 2 2	M	5 A
PRODUCT FAMILY	SIZE	PACKAGE CODE	INDUCTANCE VALUE	TOLERANCE	SERIES
		ER = tape and reel	R22 = 0.22 μ H	M = 20 %	

PERFORMANCE GRAPHS

PERFORMANCE GRAPHS: INDUCTANCE AND Q VS. FREQUENCY




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